

Chapter 1

Project Planning & Management

Project Proposal

1. Overview of the Project

The proposed project is an **AI-Powered Study Platform** designed to enhance the learning experience through intelligent educational assistance, personalized study support, and accessibility-focused features. The system is based on an **Agentic AI architecture integrated with Retrieval-Augmented Generation (RAG)** to provide context-aware and adaptive responses.

Unlike traditional educational systems, the platform is intended to understand student needs, analyze uploaded learning materials, generate summaries, create quizzes, recommend study plans, and provide step-by-step explanations. It also supports students with different learning preferences and accessibility requirements through inclusive design features such as voice narration, simplified explanations, and distraction-free study modes.

The platform targets school students, university students, lifelong learners, and users who require flexible or accessible learning environments.

2. Project Objectives

The main objectives of the proposed system are:

- To build an intelligent learning platform powered by modern AI technologies.
- To assist students in understanding educational content through summaries and explanations.
- To generate adaptive quizzes and automated feedback based on user performance.
- To provide personalized study plans according to learner goals and progress.
- To support conversational AI interaction for academic assistance.
- To implement RAG-based retrieval for accurate responses from uploaded materials.

- To improve accessibility through voice support, high-contrast UI, and simplified content modes.
- To create a scalable web platform with secure user management and cloud-based storage.
- To integrate collaboration features such as study rooms and shared learning spaces.

3. Project Scope

The scope of the project includes the design, development, integration, testing, and deployment of a complete AI-based study platform.

Included Modules

Module	Description
User Management	Registration, login, profiles, authentication
Content Upload	Upload PDFs, notes, and text materials
AI Chat Assistant	Conversational academic support
Smart Summarization	Structured and simplified summaries
Quiz Generator	Adaptive quizzes with automatic correction
Study Plan Generator	Personalized weekly learning roadmap
RAG Retrieval Engine	Search and retrieve relevant learning content
Memory & Personalization	Context-aware responses using user history
Accessibility Features	Text-to-speech, large fonts, high contrast mode

Technical Scope

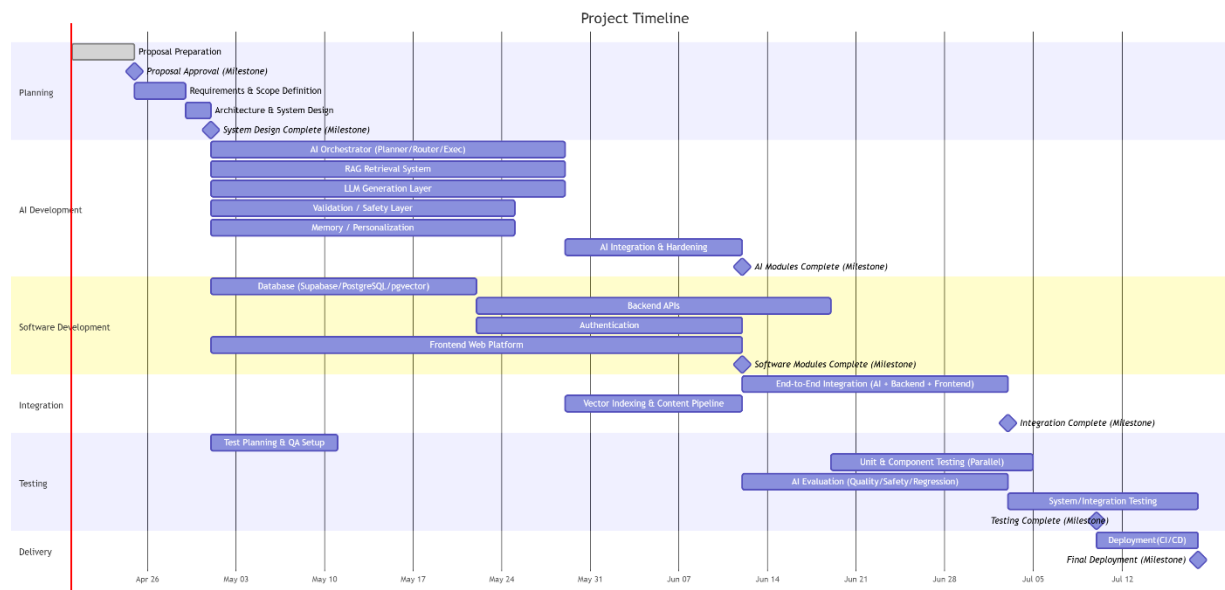
- Frontend web application using modern UI technologies.
- Backend services with database, authentication, and storage.
- AI layer using LLMs, embeddings, and orchestration modules.
- Testing, optimization, deployment, and documentation.

Limitations

- Initial deployment may depend on third-party AI API quotas or costs.
- Real-time collaboration features may be limited in first release.
- AI responses may require validation to reduce hallucinations.
- Mobile-native applications are outside the first project scope.

Project Plan

1. Timeline (Gantt Chart)



Milestones and Deliverables

Milestone 1: Discovery and Strategy

Goal: Define the project scope, business requirements, technical stack, and overall architecture of the AI-powered study platform.

Deliverables:

- **Project Requirements Specification:** Complete definition of functional and non-functional requirements, user roles, learning scenarios, and accessibility needs.

- **System Architecture Blueprint:** High-level design showing interaction between Frontend, Backend, Database, AI modules, and external APIs.
 - **Technology Stack Selection:** Final selection of frameworks such as React, FastAPI, Supabase, LangChain, and LLM providers.
 - **Project Timeline & Team Allocation:** Work breakdown structure, milestones, and responsibilities for all 10 team members.
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Milestone 2: Design and Core Platform Setup

Goal: Establish the platform foundation including UI/UX design, database structure, and backend environment.

Deliverables:

- **UI/UX Wireframes & Mockups:** Interfaces for Dashboard, Upload Page, Chat Assistant, Quiz Module, and User Profile.
 - **Database Schema Design:** Tables for users, documents, quizzes, sessions, progress tracking, and embeddings storage.
 - **Supabase Environment Setup:** Authentication, storage buckets, PostgreSQL configuration, and pgvector activation.
 - **Backend API Skeleton:** Initial APIs for authentication, uploads, and user management.
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Milestone 3: AI Core and Infrastructure (Alpha)

Goal: Build the core AI engine using Agentic AI + RAG architecture.

Deliverables:

- **AI Orchestrator Module:** Planner, router, task executor, and response merger logic.
- **RAG Retrieval Pipeline:** Document chunking, embeddings generation, vector search, reranking, and context retrieval.
- **LLM Generation Layer:** Prompt builder, model routing, response generation, and formatting.
- **Memory System:** Chat history storage, summarization memory, and personalization context.

- **Basic AI Chat Prototype:** Functional chatbot capable of answering from uploaded study materials.
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Milestone 4: Smart Features and Full Integration (Beta)

Goal: Integrate AI modules with software modules and implement advanced educational features.

Deliverables:

- **Smart Summarization System:** Automatic summaries, simplified explanations, and key-point extraction.
 - **Quiz Generation Engine:** MCQ quizzes, short-answer questions, grading, and feedback generation.
 - **Frontend + Backend + AI Integration:** Real-time communication between UI, APIs, and AI services.
 - **Validation & Safety Layer:** Hallucination checks, output verification, confidence scoring, and fallback handling.
 - **Accessibility Features:** High contrast mode, responsive UI, keyboard navigation, and assistive options.
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Milestone 5: Testing, Deployment, and Final Release (Launch)

Goal: Ensure quality, optimize performance, and deploy the production-ready platform.

Deliverables:

- **Comprehensive Testing & QA:** Unit testing, integration testing, AI evaluation, regression testing, and UAT.
- **Performance Optimization:** Faster retrieval, reduced latency, caching improvements, and API efficiency.
- **Production Deployment:** Live frontend, backend, database, and secured cloud environment.
- **Technical Documentation:** API docs, architecture report, database schema, setup guide, and maintenance notes.

- **Final Presentation:** Demo slides, live walkthrough, and project defense materials.

4. Resource Allocation

Team Distribution

Team	Members	Responsibilities
AI Engineering Team	5	AI orchestration, RAG, generation, safety, personalization
Software Engineering Team	5	Frontend, backend, database, testing, deployment

Detailed Allocation

Team Member Allocation

Member	Role	Main Responsibilities
AI Engineer 1	AI Orchestrator Engineer	Intent detection, task routing, multi-step execution, AI workflow management
AI Engineer 2	Retrieval Engineer	Document processing, chunking, embeddings, hybrid search, pgvector
AI Engineer 3	AI Generation Engineer	Chat backend, summaries, quiz engine, study plan generation
AI Engineer 4	Validation Engineer	Output validation, hallucination checks, scoring, AI quality control
AI Engineer 5	Memory Engineer	Chat memory, personalization, context storage, adaptive responses
Software Engineer 1	Frontend Engineer	Home page, navigation, landing pages, responsive layout
Software Engineer 2	Frontend Engineer	Dashboard, analytics, profile page, settings pages

Software Engineer 3	Frontend Engineer	Chat UI, quiz pages, summaries interface, upload pages
Software Engineer 4	Frontend + Integration Engineer	Frontend API connection, auth pages, state management
Software Engineer 5	Backend + DevOps Engineer	Authentication backend, deployment, CI/CD, monitoring

Technical Work Distribution

Team / Area	Members	Main Responsibilities
AI Team (AI Backend)	5 AI Engineers	Chat backend, Agentic orchestration, RAG pipeline, embeddings, pgvector management, summarization engine, quiz engine, memory system, AI backend APIs, Testing of AI modules, LangSmith tracking, evaluation metrics, prompt monitoring, model optimization
Software Team (Platform Team)	5 Software Engineers	Frontend development, authentication system, user management, core backend APIs, dashboard, profile pages, upload modules, system testing, UI testing, CI/CD pipeline, hosting, deployment, monitoring
Shared Responsibilities (Joint Work)	AI Team + Software Team	Supabase integration, PostgreSQL schema, API connection, security rules, end-to-end testing, bug fixing, production deployment, performance optimization, final documentation

Risk Assessment & Mitigation Plan

Effective risk management is essential to ensure project success, maintain development quality, control costs, and deliver a stable AI-powered platform on schedule. The following table summarizes the major identified risks, their expected impact, and the planned mitigation strategies.

Risk Category	Identified Risk	Impact Level	Mitigation Strategy
AI Reliability	LLM hallucinations or responses not grounded in uploaded study materials	High	Implement validation layer, RAG grounding, confidence scoring, and fallback responses when evidence is insufficient
AI Quality	Low-quality summaries, quizzes, or explanations that reduce educational value	High	Continuous prompt tuning, benchmark testing, user feedback loops, and output evaluation metrics
Financial / Operational	API cost overruns or rate limiting due to heavy LLM usage	High	Tiered model routing, semantic caching, request limits, token monitoring, and fallback providers such as Groq
System Performance	High response latency caused by multi-agent workflows or retrieval delays	Medium	Use asynchronous execution, parallel processing, streaming responses, optimized prompts, and caching
Data Quality	Poor parsing of PDFs, scanned files, tables, or badly formatted documents	Medium	Build structure-aware ingestion pipeline, OCR fallback if needed, file validation, and chunk quality checks
Retrieval Accuracy	Relevant content not retrieved from vector database	High	Hybrid search (vector + keyword), reranking, query rewriting, chunk optimization, pgvector indexing

Integration	Misalignment between React frontend, FastAPI backend, and AI APIs	Medium	Define API contracts early, use JSON schemas, modular endpoints, and weekly integration testing
Database Performance	Slow queries or scaling issues with embeddings and metadata storage	Medium	Optimize PostgreSQL indexes, pgvector tuning, query caching, and schema normalization
Security	Unauthorized access, weak authentication, or data exposure	High	Use Supabase Auth, rolebased access control, secure storage rules, environment variables, and input validation
Deployment	Production deployment failures or unstable releases	Medium	CI/CD pipeline, staging environment, rollback plan, health monitoring, and prerelease testing
Testing / QA	Hidden bugs discovered late in development	High	Continuous unit testing, integration testing, regression testing, and user acceptance testing
Timeline Management	Delays in milestones due to underestimation or blockers	Medium	Weekly sprint reviews, task prioritization, parallel execution, and contingency buffers
Team Management	Uneven workload distribution or dependency bottlenecks	Medium	Balanced task allocation, progress tracking, and rapid reassignment when necessary
Communication	Miscommunication between AI and software sub-teams	Medium	Weekly sync meetings, shared documentation, version control workflow, and task board updates

User Experience	Complex interface reduces usability for students	Medium	UI testing, iterative UX improvements, accessibility review, and responsive design validation
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Risk Monitoring Strategy

Monitoring Area	Action Plan
Weekly Progress Review	Track milestones, blockers, and workload balance
Cost Monitoring	Review API usage, token consumption, and provider limits
AI Evaluation	Measure answer quality, hallucination rate, and relevance
System Monitoring	Observe latency, uptime, and backend performance
Testing Control	Maintain bug tracker and regression checklist
Release Readiness	Final stabilization checks before deployment

KPIs (Key Performance Indicators)

1. Overview

Key Performance Indicators (KPIs) provide a structured, measurable framework for evaluating project success. They ensure both **technical performance** and **educational impact** can be objectively tracked and improved throughout the project lifecycle.

The KPIs are grouped into four main categories:

- **AI Quality** – Accuracy and reliability of AI-generated outputs.
- **System Performance** – Speed, uptime, and scalability of the platform.
- **Educational Impact** – Measurable improvement in student learning outcomes.
- **User Adoption Rate** – Engagement and satisfaction of platform users.

2. AI Quality KPIs

The system uses the RAGAS framework to evaluate both the retrieval and generation stages of the AI pipeline.

Table 1: AI Quality KPIs

KPI	Definition	Target	Measurement Method	Tools
Faithfulness Score	Percentage of AI responses grounded in the uploaded document context — measures hallucination prevention.	≥ 90%	Compare AI response against retrieved context chunks using RAGAS scoring.	RAGAS, DeepEval
Answer Relevancy Score	How semantically close the AI answer is to the user's original query.	≥ 85%	Cosine similarity between query embedding and response embedding.	RAGAS, SentenceTransformers
Context Precision	Signal-to-noise ratio of retrieved document chunks — relevance of retrieved passages.	≥ 80%	RAGAS retrieval evaluation on labeled test dataset.	RAGAS
Context Recall	Completeness of retrieval — whether all information needed to answer the question is retrieved.	≥ 80%	Compare retrieved chunks against expected groundtruth passages.	RAGAS

Correct Answer Rate	Overall percentage of AI answers that are factually correct relative to the document.	$\geq 88\%$	Semantic similarity comparison with expert-labeled ground truth answers.	SentenceTransformers, LangSmith
Hallucination Rate	Percentage of responses containing information not present in uploaded documents.	$< 5\%$	Guardrails layer audit logs — Gatekeeper and Auditor checks per query.	LangSmith, DeepEval

3. System Performance KPIs

These metrics ensure the platform is fast, stable, and capable of handling real user load.

Table 2: System Performance KPIs

KPI	Definition	Target	Measurement Method	Tools
Response Latency	Time from user submitting a chat query to receiving the first AI response token.	≤ 3 sec (P95)	FastAPI middleware timer logging request-to-first token duration.	FastAPI Middleware, LangSmith
Document Ingestion Time	Time to fully process an uploaded document through parsing, chunking, and embedding.	≤ 60 sec per 20-page PDF	Pipeline start timestamp to final chunk stored in Supabase.	FastAPI Logs, Supabase
System Uptime	Percentage of time the platform is fully operational and accessible.	$\geq 99\%$	Continuous healthcheck endpoint monitoring.	Cloud Hosting Monitor

Throughput	Number of concurrent API requests the system handles while maintaining target latency.	≥ 50 concurrent users	Load testing with simulated concurrent user sessions.	Locust
Cost per Request	Average LLM token cost per user chat interaction (retrieval + generation tokens).	$\leq \$0.01$ per query	Token count $\times$ model pricing rate, logged per API call.	API Logs (Groq / OpenAI)

4. Educational Impact KPIs

These are the core focus KPIs — they measure whether the platform actually improves student learning outcomes.

Table 3: Educational Impact KPIs

KPI	Definition	Target	Measurement Method	Tools
Learning Improvement (%)	Overall knowledge gain between pre-test and post-test scores for the experimental group vs. control group.	$\geq 20\%$ improvement	Pre-test vs. Post-test score comparison. Statistical validation via independent samples t-test ($p < 0.05$).	Python (scipy.stats), Pandas
Quiz Score Improvement (%)	Average percentage improvement in quiz scores across multiple attempts per student.	$\geq 15\%$ across 3+ attempts	Track quiz attempt scores per student per document over time.	Supabase DB

Error Reduction Rate (%)	Decrease in repeated mistakes across quiz attempts — measures knowledge retention.	≥ 25% reduction by attempt 3	Compare error types per student across quiz attempts.	Backend Analytics
Quiz Completion Rate	Percentage of generated quizzes that students complete in full.	≥ 75%	Count completed submissions / total quizzes started.	Supabase DB

5. User Adoption Rate KPIs

These KPIs measure how actively users engage with the platform's core features.

Table 4: User Adoption & Engagement KPIs

KPI	Definition	Target	Measurement Method	Tools
User Adoption Rate	Percentage of registered users who actively use core AI features (chat, summary, quiz) within 7 days of registration.	≥ 70%	Count active users (≥1 AI interaction) / total registered users.	Supabase Auth Logs
Document Upload Rate	Average number of documents uploaded per active user per week.	≥ 2 docs/user/week	Total uploads / active users per time period.	Supabase Storage Logs
Session Duration	Average time a user spends in a single active study session on the platform.	≥ 15 min/session	Session start to last activity timestamp.	Supabase Realtime

User Satisfaction Score	User-reported satisfaction with AI responses and platform features via postsession survey.	≥ 4.0 / 5.0	In-app feedback widget rating after each AI interaction.	Supabase DB
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6. KPI Measurement & Reporting Plan

To ensure KPIs are consistently tracked, the following reporting cadence is established:

Table 5: Measurement & Reporting Plan

Frequency	KPIs Tracked	Reporting Method
Real-Time	Response Latency, Hallucination Rate, System Uptime	LangSmith Dashboard, FastAPI Logs
Per Session	Quiz Completion Rate, Session Duration, Error Reduction Rate	Supabase DB auto-logging
Weekly	Faithfulness Score, Context Precision, Context Recall, User Adoption Rate	RAGAS aggregated report + team review
Per Milestone	Correct Answer Rate, Learning Improvement, Cost per Request	Milestone review document
End of Project	Learning Improvement (t-test), User Satisfaction Score	Final evaluation report with statistical analysis